

Cost and latency: prompt-based LLM judges vs a trained CLM classifier

Worked example: the 5-class regulatory screener (one decision per chat message)

CLASP - estimates as of 2026-10-10
Volume scenario: 1,000,000 classifications / day

At this volume, the trained CLM costs about **\$1.2K a month** on 2 rented GPUs. A prompt-based judge costs **\$5.2K to \$225.6K** a month, depending on the model. The CLM answers in **80 ms**, while the judges take **566 ms to 4.6 s**. The CLM runs one encoder pass and generates nothing, so its cost does not depend on prompt length or reasoning tokens.

